

CS 631 - Project

Deliverable 1

Requirements

Deliverable 1 must include an analysis of the intended database system and a conceptual schema. It should describe any challenges encountered during this phase and provide justification for the design decisions made.

What is expected:

1. Outline the goals of this phase of the project.
2. Analyze the entire application for the Research Lab Manager and develop an extended ER diagram. Use the notation from Chapter 3 of the textbook, along with additional notation for specialization/generalization hierarchies as presented in Chapter 4 or the course slides.
 - Identify and represent entity types, relationship types (including class/subclass relationships), and attributes.
 - Include various attribute types as needed: simple/composite, single-valued/multi-valued, stored/derived.
 - Show roles in recursive relationships.
 - Clearly indicate key attributes.
 - Include structural constraints: cardinality ratios and participation constraints. Use both the traditional notation and the (min, max) notation as an exercise.
 - Use a diagram editor to produce a clean, readable ER diagram.
3. Document any assumptions made beyond the provided project requirements. Pay particular attention to assumptions that influenced your determination of structural constraints.
4. List additional constraints that cannot be directly represented in the ER diagram. This includes any extra keys for entity types, if applicable.

NOTE: The above specification will drive your subsequent phases. As in any actual design and implementation, you will be revising what you do here as you proceed through the phases. Please be explicit and detailed when writing your deliverable. This will help you in the long run!